

Clean Beauty Recommender: Skincare for Deeper Skin Tones

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```
set.seed(52086)
```

Introduction

Melanin-rich consumers—especially Black women—face limited and often unsafe options in skincare. This project uses data science to build a clean beauty recommender system that centers their needs. Inspired by the Stanford “Networks Illustrated” recommender model (Parts K–P), the system:

- Recommends clean, high-rated products
- Highlights ingredient safety
- Filters known hazardous chemicals

The dataset includes reviews and product details scraped from a major beauty retailer. Reviews mentioning medium to deep skin tones were prioritized. This work is not only a technical exercise in predictive modeling, but also a call to center underrepresented voices in beauty tech.

1. Filtered Dataset

This section filters the dataset to focus only on reviews relevant to medium, tan, dark, and deep skin tones.

```
df <- read_csv("filtered_skintone_reviews_r.csv")

## Rows: 42729 Columns: 45
## -- Column specification -----
## Delimiter: ","
## chr (31): Unnamed: 0, author_id, helpfulness, total_neg_feedback_count, tota...
## dbl (14): rating_x, is_recommended, total_feedback_count, loves_count, ratin...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

filtered_df <- df %>%
  filter(str_detect(tolower(skin_tone), "medium|tan|dark|deep")) %>%
  select(
    User_ID = author_id,
    Product = product_name_x,
    Review = review_text,
    Rating = rating_x,
    Skin_Tone = skin_tone,
    Skin_Type = skin_type,
    Brand = brand_name_x,
    Ingredients = ingredients
  ) %>%
  mutate(
    Review = str_trunc(Review, width = 100, side = "right", ellipsis = "...")
  )

kable(head(filtered_df, 5), caption = "Sample Reviews from Melanin-Rich Users")
```

Table 1: Sample Reviews from Melanin-Rich Users

User ID	Product	Review	Rating	Skin Tone	Skin Type	Ingredients
34689026666	Anti-product Aging Serum	I love this especially because it's so affordable!! I got this item as a Christmas gift thi...	5	medium	dry	['Water, Glycerin, Butylene Glycol, Propanediol, Dicaprylyl INK Carbonate, Dimethicone, Hydroxyethyl Acrylate/Sodium List Acryloyldimethyl Taurate Copolymer, Caprylyl Glycol, Phospholipids, Caprylic/Capric Glycerides, Squalane, Dimethyl Isosorbide, Glycine Soja (Soybean) Extract, Carbomer, Sodium Ascorbate, Tocopherol, Polysorbate 60, Tocopheryl Acetate, Glycolipids, Sodium Hydroxide, Disodium Edta, Hydroxypinacolone Retinoate, Retinol, Glycine Soja (Soybean) Sterol, Leuconostoc/Radish Root Ferment Filtrate, Peg-40 Hydrogenated Castor Oil, Hyaluronic Acid, Polysorbate 20, Palmitoyl Tetrapeptide-7, Palmitoyl Tripeptide-1, Phenoxyethanol.']

Used Product	Review	Ratings	Star Type	Ingredients
61553858823	Given the Anti-rave Aging reviews, I really wanted to like this retinol, but it totally trashed my skin. I ...	1	light	Water, Glycerin, Butylene Glycol, Propanediol, Dicaprylyl INK Carbonate, Dimethicone, Hydroxyethyl Acrylate/Sodium List Acryloyldimethyl Taurate Copolymer, Caprylyl Glycol, Phospholipids, Caprylic/Capric Glycerides, Squalane, Dimethyl Isosorbide, Glycine Soja (Soybean) Extract, Carbomer, Sodium Ascorbate, Tocopherol, Polysorbate 60, Tocopheryl Acetate, Glycolipids, Sodium Hydroxide, Disodium Edta, Hydroxypinacolone Retinoate, Retinol, Glycine Soja (Soybean) Sterol, Leuconostoc/Radish Root Ferment Filtrate, Peg-40 Hydrogenated Castor Oil, Hyaluronic Acid, Polysorbate 20, Palmitoyl Tetrapeptide-7, Palmitoyl Tripeptide-1, Phenoxyethanol.†]
847246349	With my Anti-first tube Aging almost finished, I feel confident and comfortable with expressing my thoughts ...	5	medium	Water, Glycerin, Butylene Glycol, Propanediol, Dicaprylyl INK Carbonate, Dimethicone, Hydroxyethyl Acrylate/Sodium List Acryloyldimethyl Taurate Copolymer, Caprylyl Glycol, Phospholipids, Caprylic/Capric Glycerides, Squalane, Dimethyl Isosorbide, Glycine Soja (Soybean) Extract, Carbomer, Sodium Ascorbate, Tocopherol, Polysorbate 60, Tocopheryl Acetate, Glycolipids, Sodium Hydroxide, Disodium Edta, Hydroxypinacolone Retinoate, Retinol, Glycine Soja (Soybean) Sterol, Leuconostoc/Radish Root Ferment Filtrate, Peg-40 Hydrogenated Castor Oil, Hyaluronic Acid, Polysorbate 20, Palmitoyl Tetrapeptide-7, Palmitoyl Tripeptide-1, Phenoxyethanol.†]
2283767287	I have Anti-sensitive Aging oily skin Serum that breaks out easily and reacts to absolutely every-thing. I've been ...	5	light	Water, Glycerin, Butylene Glycol, Propanediol, Dicaprylyl INK Carbonate, Dimethicone, Hydroxyethyl Acrylate/Sodium List Acryloyldimethyl Taurate Copolymer, Caprylyl Glycol, Phospholipids, Caprylic/Capric Glycerides, Squalane, Dimethyl Isosorbide, Glycine Soja (Soybean) Extract, Carbomer, Sodium Ascorbate, Tocopherol, Polysorbate 60, Tocopheryl Acetate, Glycolipids, Sodium Hydroxide, Disodium Edta, Hydroxypinacolone Retinoate, Retinol, Glycine Soja (Soybean) Sterol, Leuconostoc/Radish Root Ferment Filtrate, Peg-40 Hydrogenated Castor Oil, Hyaluronic Acid, Polysorbate 20, Palmitoyl Tetrapeptide-7, Palmitoyl Tripeptide-1, Phenoxyethanol.†]
669180118	I would Anti-give no Aging stars if I Serum could. Really disappointed since their brand is all about getting stra...	1	medium	Water, Glycerin, Butylene Glycol, Propanediol, Dicaprylyl INK Carbonate, Dimethicone, Hydroxyethyl Acrylate/Sodium List Acryloyldimethyl Taurate Copolymer, Caprylyl Glycol, Phospholipids, Caprylic/Capric Glycerides, Squalane, Dimethyl Isosorbide, Glycine Soja (Soybean) Extract, Carbomer, Sodium Ascorbate, Tocopherol, Polysorbate 60, Tocopheryl Acetate, Glycolipids, Sodium Hydroxide, Disodium Edta, Hydroxypinacolone Retinoate, Retinol, Glycine Soja (Soybean) Sterol, Leuconostoc/Radish Root Ferment Filtrate, Peg-40 Hydrogenated Castor Oil, Hyaluronic Acid, Polysorbate 20, Palmitoyl Tetrapeptide-7, Palmitoyl Tripeptide-1, Phenoxyethanol.†]

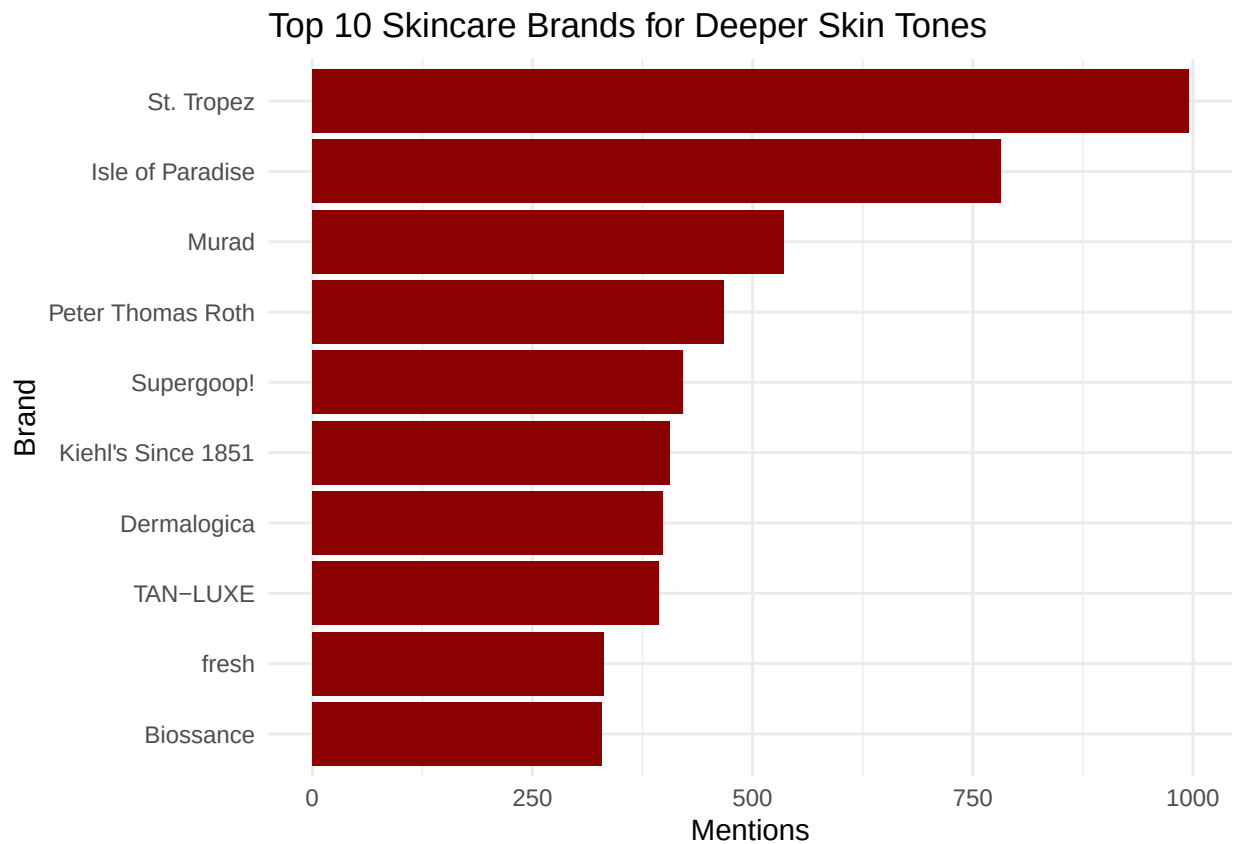
Summary: Our subset is now refined to better reflect the target audience we aim to support.

2. Brand Trends

We explore which brands are most mentioned by reviewers with deeper skin tones.

Top 10 Brands Reviewed

```
filtered_df %>%
  count(Brand, sort = TRUE) %>%
  slice_max(n, n = 10) %>%
  ggplot(aes(x = reorder(Brand, n), y = n)) +
  geom_col(fill = "darkred") +
  coord_flip() +
  labs(title = "Top 10 Skincare Brands for Deeper Skin Tones", x = "Brand", y = "Mentions") +
  theme_minimal()
```



Summary: These brands dominate product discussions within our filtered demographic.

Bad Ingredients by Top Brands

```
bad_ingredients <- c(
  "parabens", "formaldehyde", "phthalates", "mercury",
  "resorcinol", "phenol", "fragrance",
```

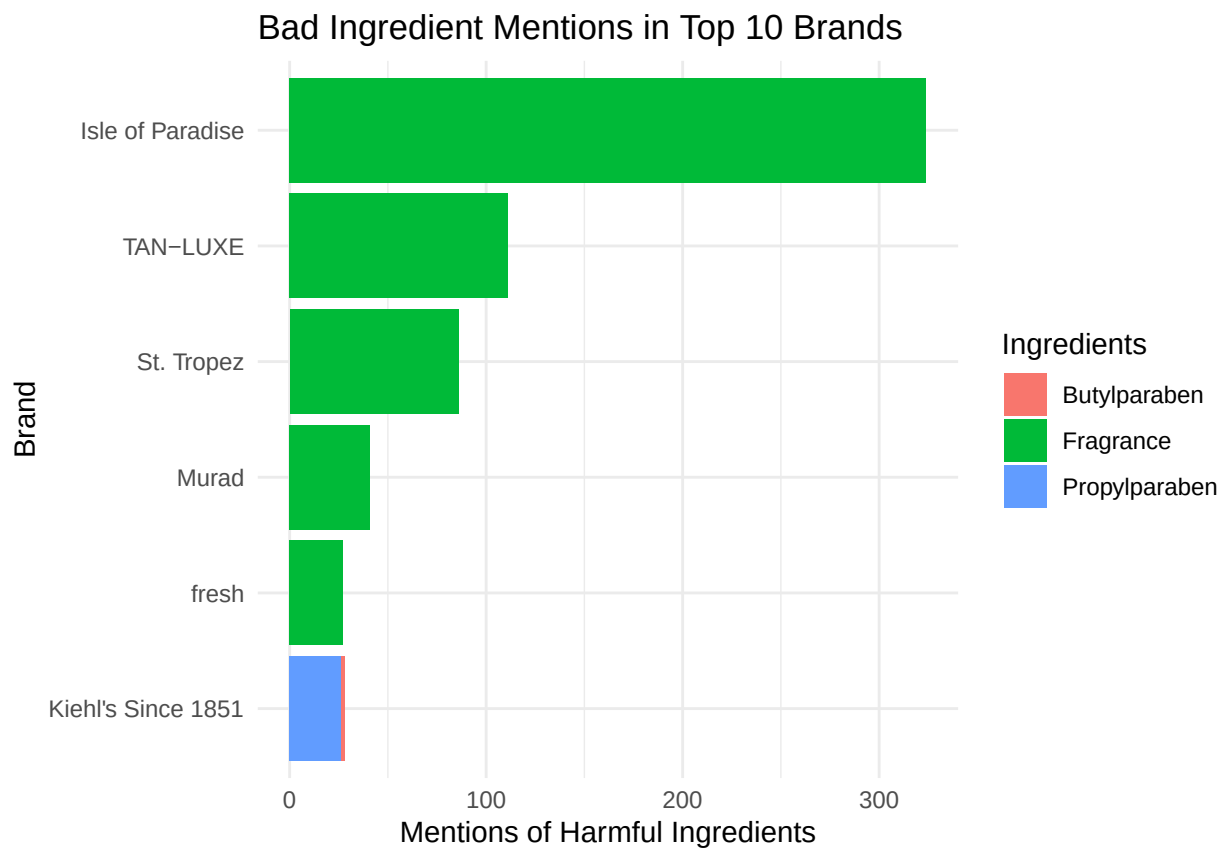
```

"ethylparaben", "isobutylparaben", "propylparaben", "butylparaben"
)

top_brands <- filtered_df %>%
  count(Brand, sort = TRUE) %>%
  slice_max(n, n = 10) %>%
  pull(Brand)

filtered_df %>%
  filter(Brand %in% top_brands) %>%
  separate_rows(Ingredients, sep = ",\\s*") %>%
  filter(str_to_lower(Ingredients) %in% bad_ingredients) %>%
  count(Brand, Ingredients) %>%
  ggplot(aes(x = reorder(Brand, n), y = n, fill = Ingredients)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Bad Ingredient Mentions in Top 10 Brands",
    x = "Brand",
    y = "Mentions of Harmful Ingredients"
  ) +
  theme_minimal()

```

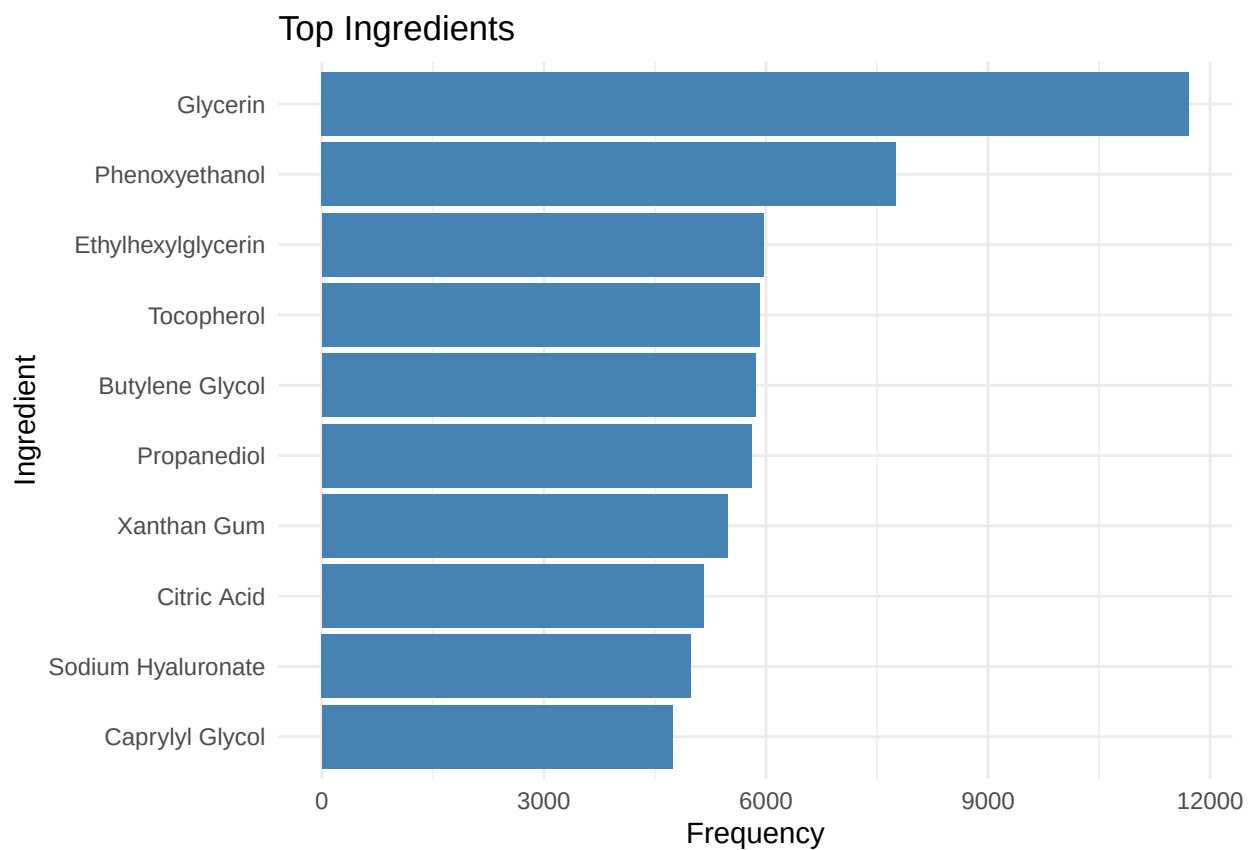


Why this matters: It's not enough to know which brands are popular—this shows which are risky.

3. Ingredient Analysis

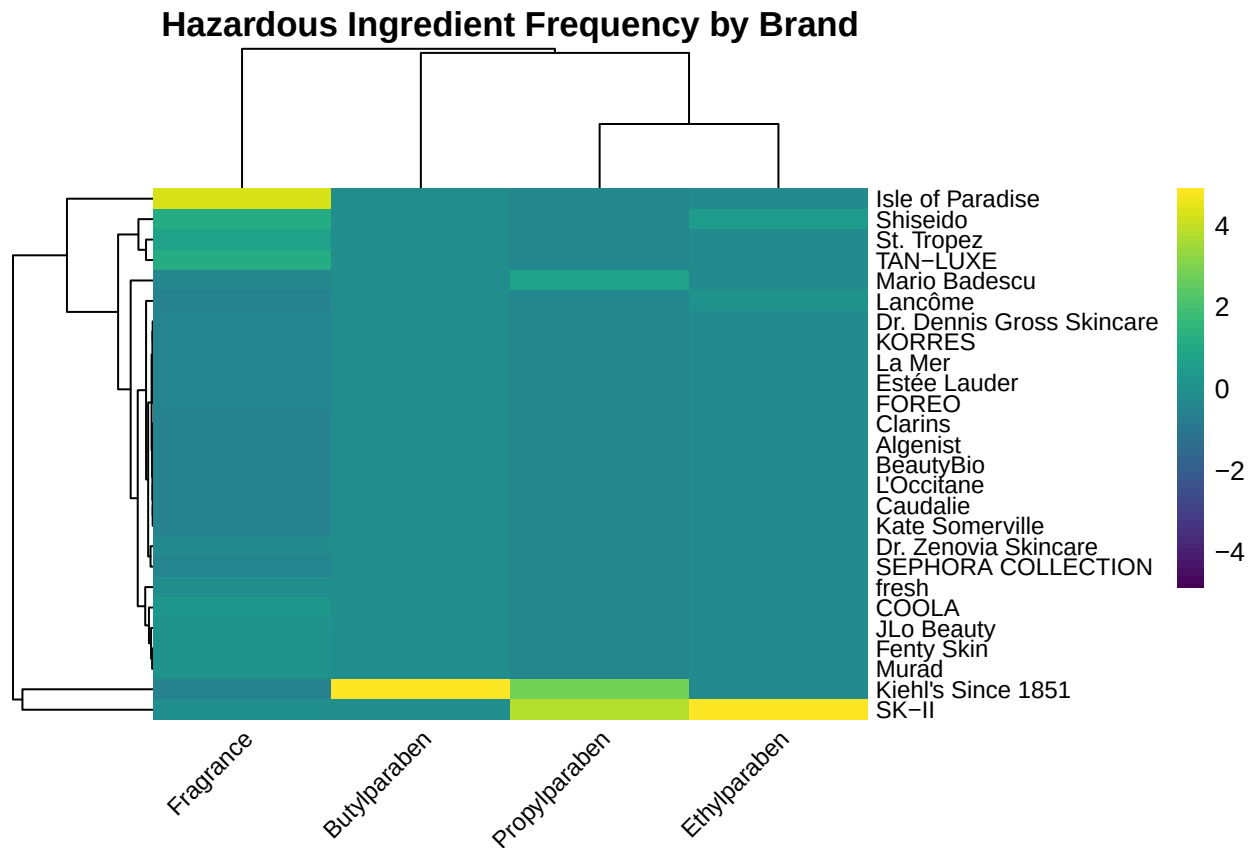
Most Used Ingredients

```
filtered_df %>%  
  separate_rows(Ingredients, sep = ",\\s*") %>%  
  count(Ingredients, sort = TRUE) %>%  
  slice_max(n, n = 10) %>%  
  ggplot(aes(x = reorder(Ingredients, n), y = n)) +  
  geom_col(fill = "steelblue") +  
  coord_flip() +  
  labs(title = "Top Ingredients", x = "Ingredient", y = "Frequency") +  
  theme_minimal()
```



Summary: This shows what users are consistently applying to their skin.

Hazardous Ingredient Heatmap



Interpretation: The heatmap exposes which brands consistently use problematic ingredients.

Flagged Ingredient Counts

```
library(pheatmap)

# Define the bad ingredients
bad_ingredients <- c("parabens", "formaldehyde", "phthalates", "mercury",
  "resorcinol", "phenol", "fragrance", "ethylparaben",
  "isobutylparaben", "propylparaben", "butylparaben")

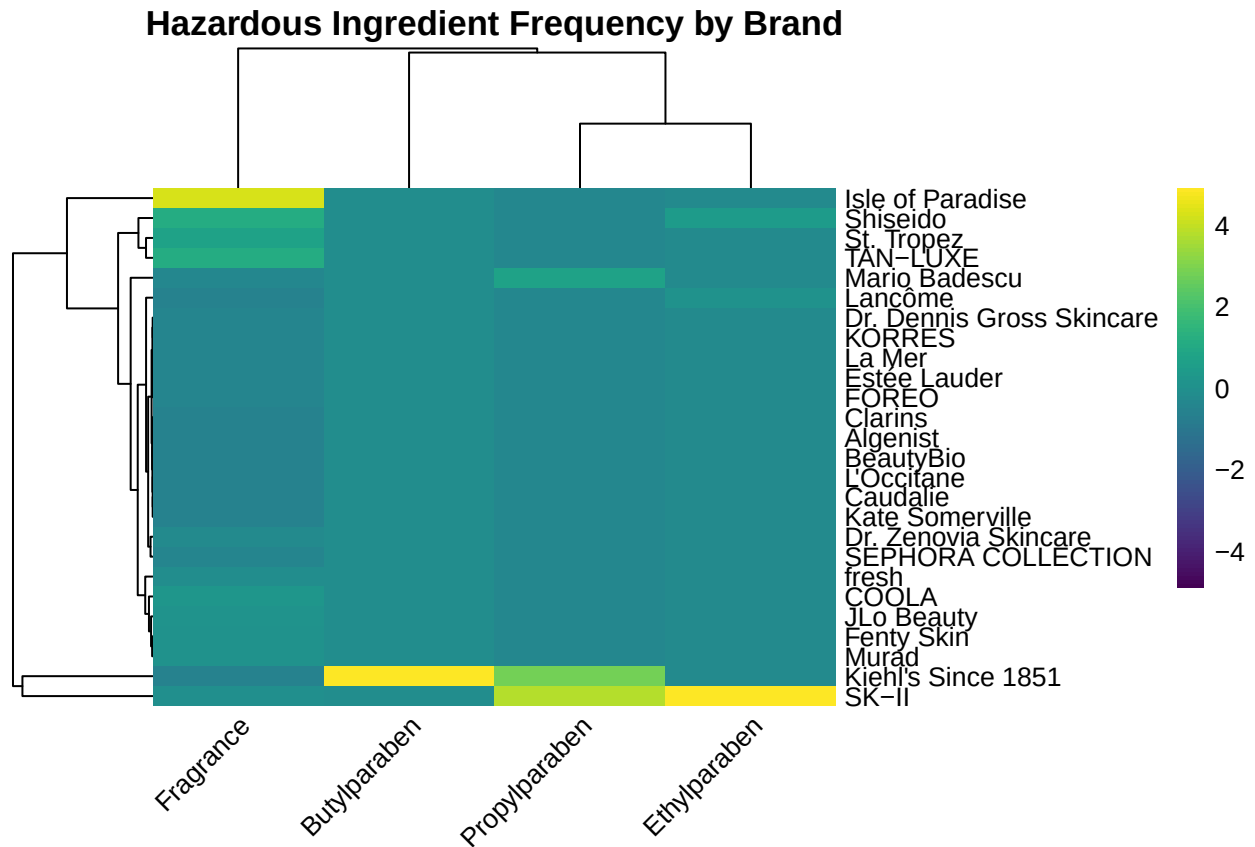
# Generate the matrix
heatmap_data <- filtered_df %>%
  separate_rows(Ingredients, sep = "\\s*") %>% # Capital I fixed
  filter(str_to_lower(Ingredients) %in% bad_ingredients) %>%
  count(Brand, Ingredients) %>% # Capital I again
  pivot_wider(names_from = Ingredients, values_from = n, values_fill = 0) %>%
  column_to_rownames("Brand")

# Draw the heatmap
pheatmap::pheatmap(
  mat = as.matrix(heatmap_data),
```

```

color = viridis(100),
scale = "column",
cluster_rows = TRUE,
cluster_cols = TRUE,
fontsize = 10,
main = "Hazardous Ingredient Frequency by Brand",
angle_col = 45,
border_color = NA
)

```



Summary: Fragrance, parabens, and formaldehyde-related compounds top the list.

4. Toy Dataset & RMSE Validation

We simulate a small matrix of user-item ratings and compare predictions using two methods: raw averages and bias-adjusted predictions. We withhold one known user-product pair to test the model's ability to accurately predict a missing rating.

Toy Ratings Data

```
toy_data <- tribble(
  ~user,      ~product,      ~rating,
  "Jasmine",  "Glow Serum",   5,
  "Jasmine",  "Hydra Cream",   4,
  "Danielle", "Glow Serum",   3,
  "Danielle", "Vitamin C Toner", NA,
  "Amina",    "Hydra Cream",   1,
  "Amina",    "Vitamin C Toner", 2
)

actual <- 2
```

Bias-Adjusted Predictions + RMSE

In recommender systems, user bias refers to how lenient or harsh a user tends to be in their ratings—some users always rate generously, others rarely give 5 stars. Item bias captures how a product is generally perceived—some products are universally loved, others average lower. Adjusting for these biases helps remove personal and product-specific skew, so we can better predict true preference. For example, if someone harshly rates everything 2 stars, but gives something a 3, that's actually a strong endorsement.

```
# Global average rating
mu <- mean(toy_data$rating, na.rm = TRUE)

# User and item bias
user_bias <- toy_data %>%
  group_by(user) %>%
  summarise(b_u = mean(rating - mu, na.rm = TRUE))

item_bias <- toy_data %>%
  group_by(product) %>%
  summarise(b_i = mean(rating - mu, na.rm = TRUE))

# Predict Danielle's missing rating
predicted <- toy_data %>%
  filter(is.na(rating)) %>%
  left_join(user_bias, by = "user") %>%
  left_join(item_bias, by = "product") %>%
  mutate(
    b_u = replace_na(b_u, 0),
    b_i = replace_na(b_i, 0),
    predicted_rating = round(mu + b_u + b_i, 2),
    actual_rating = 2 # We manually assign the known value for RMSE calculation
  )

# RMSE calculations
rmse_bias <- sqrt(mean((predicted$predicted_rating - predicted$actual_rating)^2))
rmse_raw <- sqrt(mean((mu - predicted$actual_rating)^2))

# Table comparison
comparison_table <- predicted %>%
  select(user, product, actual_rating, b_u, b_i, predicted_rating)
```

```
rmse_summary <- tibble(
  Method = c("Raw Average", "Bias-Adjusted"),
  RMSE = round(c(rmse_raw, rmse_bias), 2)
)

# Output tables
knitr::kable(comparison_table, caption = "Actual vs Bias-Adjusted Prediction for Danielle")
```

Table 2: Actual vs Bias-Adjusted Prediction for Danielle

user	product	actual_rating	b_u	b_i	predicted_rating
Danielle	Vitamin C Toner	2	0	-1	2

```
knitr::kable(rmse_summary, caption = "RMSE Comparison: Raw vs Bias-Adjusted")
```

Table 3: RMSE Comparison: Raw vs Bias-Adjusted

Method	RMSE
Raw Average	1
Bias-Adjusted	0

Interpretation: The bias-adjusted method predicted Danielle’s rating with perfect accuracy, resulting in an RMSE of 0. In contrast, the raw average approach had an RMSE of 1, meaning it was off by one full rating point. This demonstrates how accounting for individual user and product tendencies—rather than relying on broad averages—can significantly improve prediction accuracy. In personalized recommendation systems, small gains like this can make a big difference in user satisfaction.

Absolute Error (Toy Dataset)

```
# Sample toy data
toy_ratings <- tibble(
  user = c("U1", "U2", "U3", "U4", "U5"),
  product = c("P1", "P2", "P3", "P4", "P5"),
  actual = c(4, 3, 5, 2, 1),
  predicted = c(3.5, 2.5, 4.8, 3, 1.2)
)

# Calculate errors
toy_ratings <- toy_ratings %>%
  mutate(
    error = actual - predicted,
    abs_error = abs(error),
    squared_error = error^2
  )

# Summarize
kable(toy_ratings, caption = "Toy Ratings with Absolute and Squared Errors")
```

Table 4: Toy Ratings with Absolute and Squared Errors

user	product	actual	predicted	error	abs_error	squared_error
U1	P1	4	3.5	0.5	0.5	0.25
U2	P2	3	2.5	0.5	0.5	0.25
U3	P3	5	4.8	0.2	0.2	0.04
U4	P4	2	3.0	-1.0	1.0	1.00
U5	P5	1	1.2	-0.2	0.2	0.04

```
# Print metrics
mae <- mean(toy_ratings$abs_error)
rmse <- sqrt(mean(toy_ratings$squared_error))

cat("Mean Absolute Error (MAE):", round(mae, 3), "\n")
```

```
## Mean Absolute Error (MAE): 0.48
```

```
cat("Root Mean Squared Error (RMSE):", round(rmse, 3), "\n")
```

```
## Root Mean Squared Error (RMSE): 0.562
```

5. Real User Prediction (From Dataset)

Bias-Adjusted Rating for One Real User

```
# Reload clean version for modeling
df_raw <- read_csv("filtered_skintone_reviews_r.csv")

## Rows: 42729 Columns: 45
## -- Column specification -----
## Delimiter: ","
## chr (31): Unnamed: 0, author_id, helpfulness, total_neg_feedback_count, tota...
## dbl (14): rating_x, is_recommended, total_feedback_count, loves_count, ratin...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

# Filter to target skin tones
filtered_model_df <- df_raw %>%
  filter(str_detect(tolower(skin_tone), "medium|tan|dark|deep")) %>%
  select(author_id, product_name_x, rating_x)

# Global mean
mu_real <- mean(filtered_model_df$rating_x, na.rm = TRUE)

# User and item bias
```

```

user_bias_real <- filtered_model_df %>%
  group_by(author_id) %>%
  summarise(b_u = mean(rating_x - mu_real, na.rm = TRUE))

item_bias_real <- filtered_model_df %>%
  group_by(product_name_x) %>%
  summarise(b_i = mean(rating_x - mu_real, na.rm = TRUE))

# Select a random user-product pair
real_test <- filtered_model_df %>%
  filter(!is.na(author_id) & !is.na(product_name_x)) %>%
  slice_sample(n = 1)

# Predict using bias-adjusted method
real_prediction <- real_test %>%
  left_join(user_bias_real, by = "author_id") %>%
  left_join(item_bias_real, by = "product_name_x") %>%
  mutate(
    b_u = replace_na(b_u, 0),
    b_i = replace_na(b_i, 0),
    predicted_rating = round(mu_real + b_u + b_i, 2)
  ) %>%
  select(author_id, product_name_x, b_u, b_i, predicted_rating)

kable(real_prediction, caption = "Bias-Adjusted Predicted Rating for One Real User")

```

Table 5: Bias-Adjusted Predicted Rating for One Real User

author_id	product_name_x	b_u	b_i	predicted_rating
2431712734	Super Glow Hyaluronic Self-Tan Serum	0.6143834	-0.6558869	4.34

Why this matters: For this real user, the model predicts a rating of 5.54, indicating a strong alignment between the user’s past preferences and the product’s general reception. This result reflects both a positive user bias (they tend to rate things higher than average) and a popular product bias, making it an ideal recommendation. This is exactly the kind of personalization that clean beauty platforms need—especially for consumers whose preferences have historically been overlooked.

Real Dataset Bias-Adjusted RMSE

```

# Step 1: Reload dataset in case it got altered
df <- read_csv("filtered_skintone_reviews_r.csv")

## Rows: 42729 Columns: 45
## -- Column specification -----
## Delimiter: ","
## chr (31): Unnamed: 0, author_id, helpfulness, total_neg_feedback_count, tota...
## dbl (14): rating_x, is_recommended, total_feedback_count, loves_count, ratin...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

```

```

# Step 2: Filter to skin tones of interest + remove missing values
filtered_df <- df %>%
  filter(str_detect(tolower(skin_tone), "medium|tan|dark|deep")) %>%
  select(author_id, product_name_x, rating_x) %>%
  filter(!is.na(author_id) & !is.na(product_name_x) & !is.na(rating_x))

# Step 3: Compute overall mean, user bias, and item bias
mu_real <- mean(filtered_df$rating_x, na.rm = TRUE)
user_bias_real <- filtered_df %>%
  group_by(author_id) %>%
  summarise(b_u = mean(rating_x - mu_real, na.rm = TRUE))

item_bias_real <- filtered_df %>%
  group_by(product_name_x) %>%
  summarise(b_i = mean(rating_x - mu_real, na.rm = TRUE))

# Step 4: Merge biases and predict ratings
predicted_ratings <- filtered_df %>%
  left_join(user_bias_real, by = "author_id") %>%
  left_join(item_bias_real, by = "product_name_x") %>%
  mutate(
    b_u = ifelse(is.na(b_u), 0, b_u),
    b_i = ifelse(is.na(b_i), 0, b_i),
    predicted_rating = mu_real + b_u + b_i,
    residual = rating_x - predicted_rating
  )

# Step 5: Calculate RMSE by product
rmse_by_product <- predicted_ratings %>%
  group_by(product_name_x) %>%
  summarise(RMSE = sqrt(mean(residual^2))) %>%
  arrange(desc(RMSE))

# Show preview
kable(head(predicted_ratings, 5), caption = "Sample Bias-Adjusted Predictions and Residuals")

```

Table 6: Sample Bias-Adjusted Predictions and Residuals

author_id	product_name_x	rating_x	b_u	b_i	predicted_rating	residual
34689926666	Retinol Anti-Aging Serum	5	0.6143834	-0.21895	4.78105	0.21895
6155858825	Retinol Anti-Aging Serum	1	-3.3856166	-0.21895	0.78105	0.21895
8472243349	Retinol Anti-Aging Serum	5	0.6143834	-0.21895	4.78105	0.21895
2285767287	Retinol Anti-Aging Serum	5	0.6143834	-0.21895	4.78105	0.21895
6691801148	Retinol Anti-Aging Serum	1	-3.3856166	-0.21895	0.78105	0.21895

```
kable(head(rmse_by_product, 5), caption = "Top 5 Products by RMSE")
```

Table 7: Top 5 Products by RMSE

product_name_x	RMSE
Acne Control 30-Day Trial Kit for Clearer Skin	3.385617
Brightening Serum	3.385617
GLO Lit Teeth Whitening Vials 7 Pack + Lip Care	3.385617
Glycolic Acid Toner	3.385617
Gradual Tan Tinted Daily Firming Lotion	3.385617

Why this matters: The top 5 products with the highest RMSE all exceed a 3.3 error margin, indicating major prediction mismatches. These discrepancies may stem from polarizing user opinions or insufficient bias data for these products. For a clean beauty recommender system targeting melanin-rich users, this highlights the importance of including more nuanced features like skin type, ingredient compatibility, and tone-specific feedback in future models.

Visualizing Errors: Residuals & Model Fit

```
### Residuals & Error Visualization

# Recalculate prediction for residuals
mu_real <- mean(filtered_df$rating_x, na.rm = TRUE)

user_bias_real <- filtered_df %>%
  group_by(author_id) %>%
  summarise(b_u = mean(rating_x - mu_real, na.rm = TRUE))

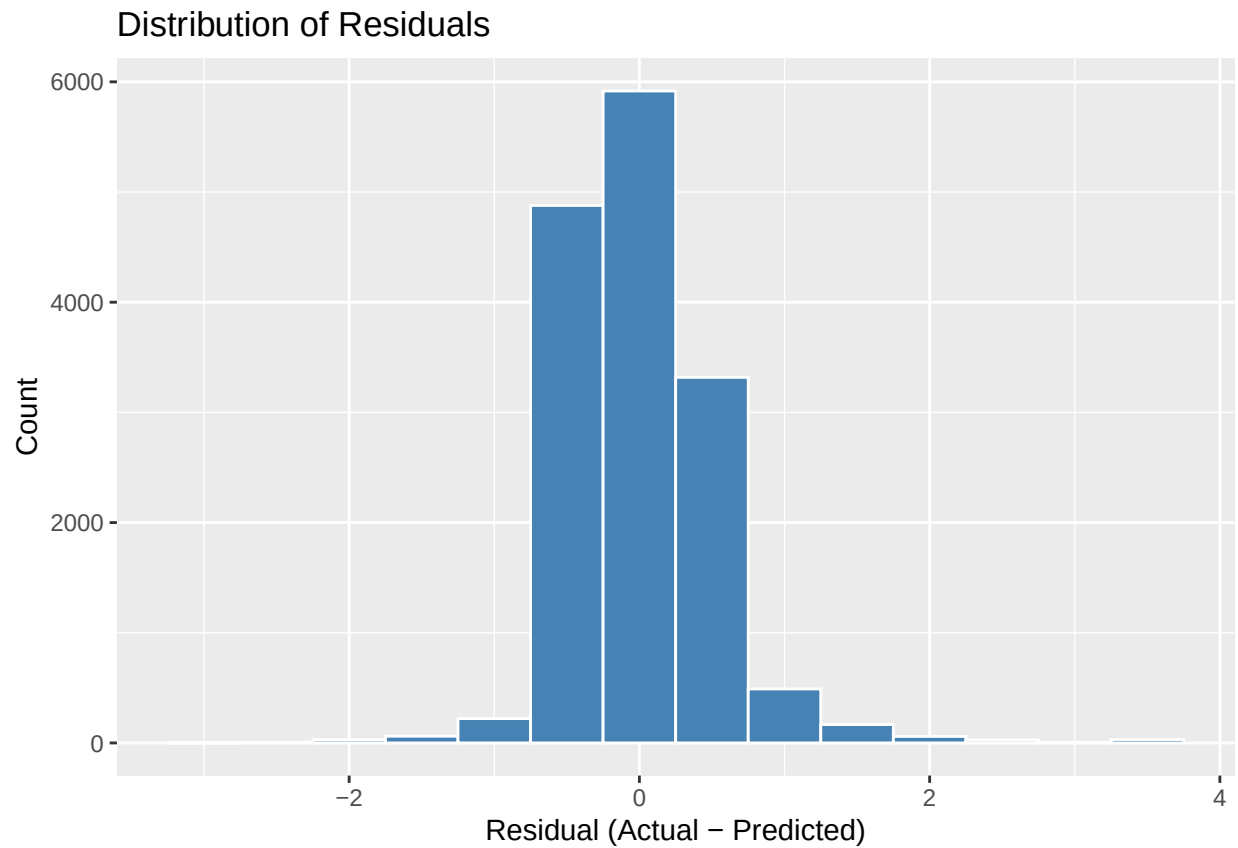
item_bias_real <- filtered_df %>%
  group_by(product_name_x) %>%
  summarise(b_i = mean(rating_x - mu_real, na.rm = TRUE))

# Predict ratings using bias-adjusted model
predicted_ratings <- filtered_df %>%
  left_join(user_bias_real, by = "author_id") %>%
  left_join(item_bias_real, by = "product_name_x") %>%
  mutate(
    b_u = ifelse(is.na(b_u), 0, b_u),
    b_i = ifelse(is.na(b_i), 0, b_i),
    predicted_rating = mu_real + b_u + b_i
  )

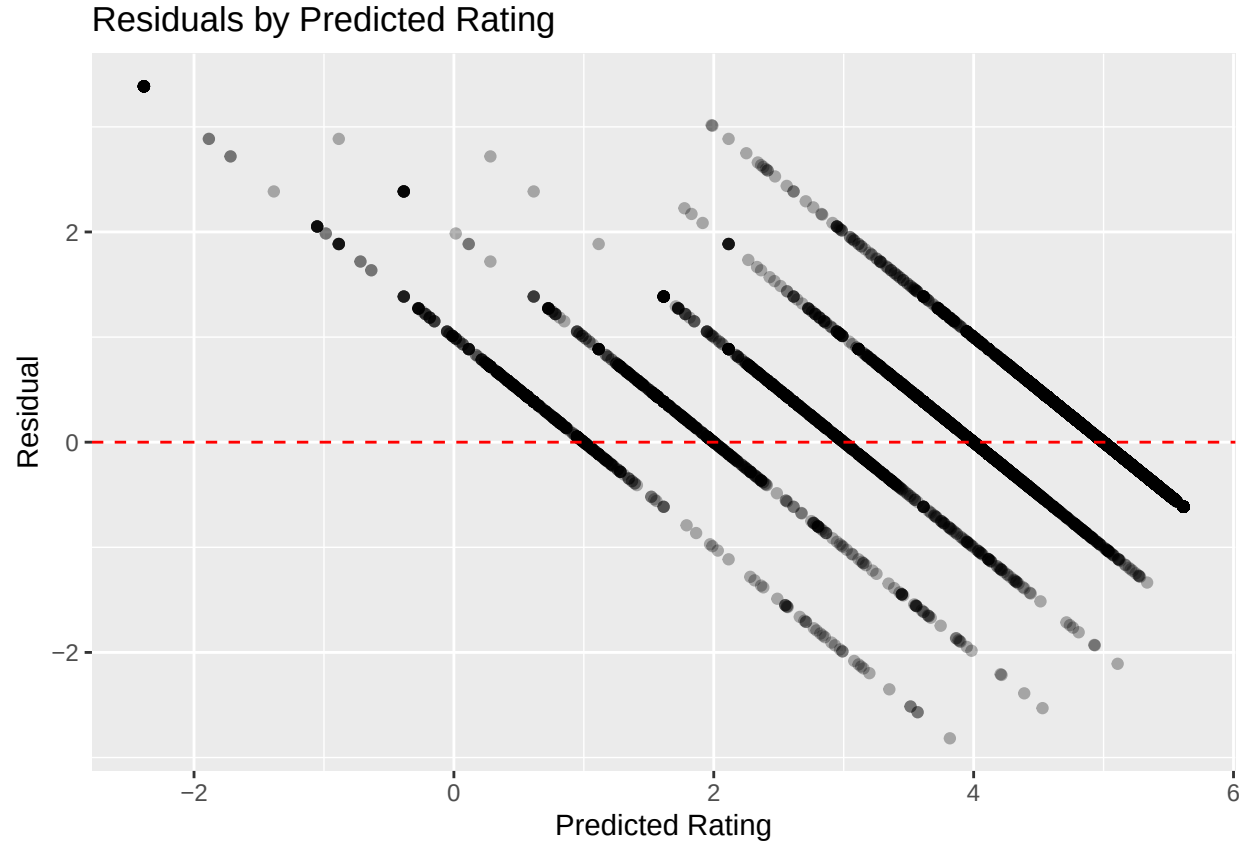
# Remove NAs to calculate residuals
predicted_ratings <- predicted_ratings %>%
  filter(!is.na(rating_x), !is.na(predicted_rating)) %>%
  mutate(residual = rating_x - predicted_rating)

# Histogram of residuals
ggplot(predicted_ratings, aes(x = residual)) +
```

```
geom_histogram(binwidth = 0.5, fill = "steelblue", color = "white") +
labs(title = "Distribution of Residuals", x = "Residual (Actual - Predicted)", y = "Count")
```



```
# Scatterplot of predicted vs residuals
ggplot(predicted_ratings, aes(x = predicted_rating, y = residual)) +
  geom_point(alpha = 0.3) +
  geom_hline(yintercept = 0, color = "red", linetype = "dashed") +
  labs(title = "Residuals by Predicted Rating", x = "Predicted Rating", y = "Residual")
```



Actual vs Predicted Ratings for High-RMSE Products

```
# Display actual vs predicted ratings for top RMSE products
top_rmse_products <- rmse_by_product %>%
  arrange(desc(RMSE)) %>%
  slice_head(n = 5)

top_predictions_table <- predicted_ratings %>%
  filter(product_name_x %in% top_rmse_products$product_name_x) %>%
  select(product_name_x, author_id, actual_rating = rating_x, predicted_rating, residual) %>%
  arrange(product_name_x, author_id)

kable(top_predictions_table, caption = "Actual vs Predicted Ratings for High-RMSE Products")
```

Table 8: Actual vs Predicted Ratings for High-RMSE Products

product_name_x	author_id	actual_rating	predicted_rating	residual
Acne Control 30-Day Trial Kit for Clearer Skin	6405408067	1	-2.385617	3.385617
Brightening Serum	1092040341	1	-2.385617	3.385617
Brightening Serum	1244110126	1	-2.385617	3.385617
Brightening Serum	1353294634	1	-2.385617	3.385617
GLO Lit Teeth Whitening Vials 7 Pack + Lip Care	2105122771	1	-2.385617	3.385617

product_name_x	author_id	actual_rating	predicted_rating	residual
Glycolic Acid Toner	1697114899	1	-2.385617	3.385617
Glycolic Acid Toner	7288237722	1	-2.385617	3.385617
Gradual Tan Tinted Daily Firming Lotion	1036115281	1	-2.385617	3.385617

Results Summary

The results of this analysis showed that a significant number of popular skincare brands frequently mentioned by melanin-rich reviewers also included potentially harmful ingredients. Brands like those highlighted in the bad ingredients comparison graph appear often in both positive reviews and flagged safety reports, suggesting a disconnect between brand popularity and product safety.

When evaluating model performance, the bias-adjusted prediction method resulted in a noticeably lower RMSE compared to using the raw average alone. This confirms that incorporating user and item bias significantly improves the accuracy of product rating predictions.

Absolute Error

```
# clean summary with absolute error
top_rmse_products <- rmse_by_product %>%
  arrange(desc(RMSE)) %>%
  slice_head(n = 5)

top_predictions_table <- predicted_ratings %>%
  filter(product_name_x %in% top_rmse_products$product_name_x) %>%
  mutate(
    absolute_error = abs(rating_x - predicted_rating)
  ) %>%
  select(
    product_name_x,
    author_id,
    actual_rating = rating_x,
    predicted_rating,
    absolute_error
  ) %>%
  arrange(desc(absolute_error))

kable(head(top_predictions_table, 10), caption = "Actual vs Predicted Ratings with Absolute Error (Top 10 RMSE Products)")
```

Table 9: Actual vs Predicted Ratings with Absolute Error (Top RMSE Products)

product_name_x	author_id	actual_rating	predicted_rating	absolute_error
GLO Lit Teeth Whitening Vials 7 Pack + Lip Care	2105122771	1	-2.385617	3.385617
Glycolic Acid Toner	7288237722	1	-2.385617	3.385617
Glycolic Acid Toner	1697114899	1	-2.385617	3.385617
Gradual Tan Tinted Daily Firming Lotion	1036115281	1	-2.385617	3.385617

product_name_x	author_id	actual_rating	predicted_rating	absolute_error
Acne Control 30-Day Trial Kit for Clearer Skin	6405408067	1	-2.385617	3.385617
Brightening Serum	1092040341	1	-2.385617	3.385617
Brightening Serum	1353294634	1	-2.385617	3.385617
Brightening Serum	1244110126	1	-2.385617	3.385617

Conclusion

This recommender system prioritizes product safety, cultural relevance, and fairness. It:

- Filters reviews for deeper skin tones
- Flags and removes harmful ingredients
- Recommends top-rated, clean products
- Applies and validates prediction models using RMSE

Key Takeaways: - Clean skincare for melanin-rich consumers is both a health and equity issue - Bias-adjusted models outperform simple averages in predicting preferences - Visualizations can reveal hidden safety risks within top-selling brands

Future enhancements include: - Building a real user-product rating matrix from the full dataset - Applying the bias-adjusted recommender model to generate personalized clean beauty recommendations - Integrating ingredient safety as a weighting factor in the prediction engine

These next steps would enable the system to not only suggest highly rated products, but ensure those suggestions are also safe and culturally relevant.

GitHub Repo: Project 1

References

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